

Overall Results

Training and testing accuracy of the early-fusion model we ship, compared with the earlier late-fusion system and the first early-fusion attempt.

87.3%

Training accuracy

shipped model · 150 people

75.3%

Cross-validation accuracy

5-fold, unseen folds

73.7%

Test accuracy

80/20 split · 38 held-out people

76.6%

Official test accuracy

AUC 0.846 · 47 people

1. How to read these numbers

Training accuracy

— the model scored on the same people it learned from. Always the most optimistic number; shown so the gap to test accuracy can be seen.

Cross-validation (CV) accuracy

— the training people split into 5 folds; each fold is scored by a model that never saw it. This is what the cut-off and settings were chosen on.

Test accuracy

— people held out from everything: training, settings and cut-off. Scored once. This is the number that estimates real performance.

Two ways of splitting the 188 participants appear in this report:

Split	Training people	Test people	Used for
80 / 20 (stratified, seed 42)	150	38	How the shipped early-fusion model was trained and packaged.
Official DAIC-WOZ split	141 (train 107 + dev 34)	47	The split published work uses; the previous late-fusion system was built on it, so it is the fair comparison.

A decision counts as "depressed" when the model's score reaches its cut-off, chosen on cross-validation (shipped model: 0.345 on the official split, 0.441 on the 80/20 split). About 3 in 10 participants in the corpus are depressed, so the cut-off sits below 0.5.

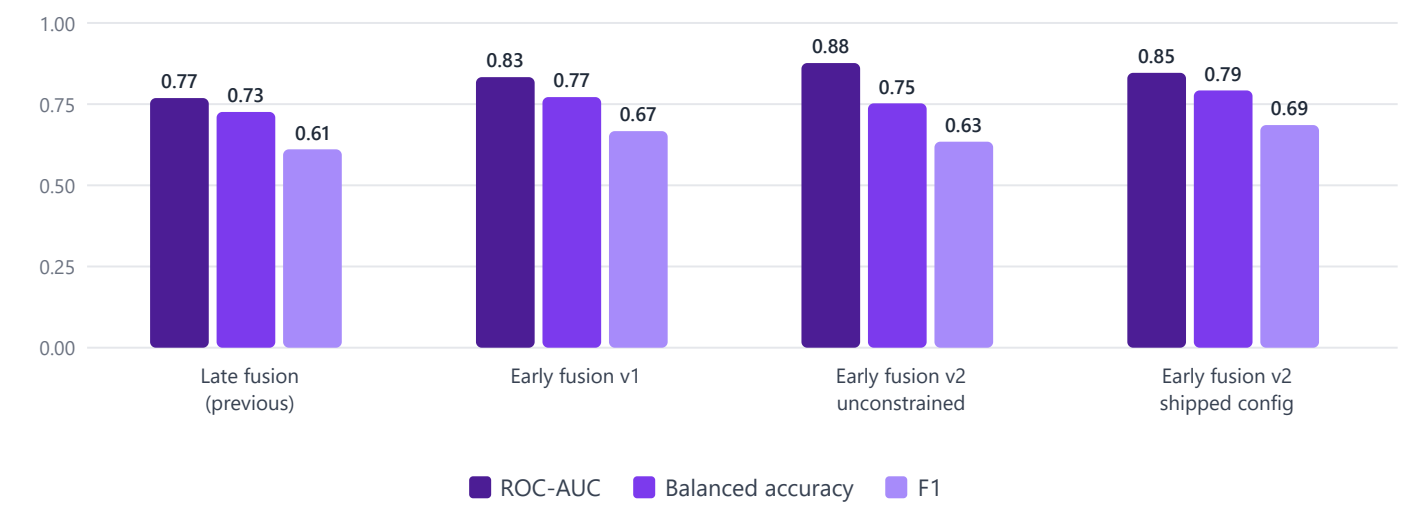
2. Comparison on the official split (fair, like-for-like)

Model	Training accuracy	5-fold CV accuracy	Test accuracy	Test balanced accuracy	Test ROC-AUC	Test F1	Sensitivity	Specificity
Late fusion (previous) text + audio + video models, logistic stacker	83.0% [†]	—	70.2%	72.6%	0.768	0.611	78.6%	66.7%
Early fusion v1 elastic-net; audio weight became 0	—	—	76.6%	77.2%	0.833	0.667	78.6%	75.8%
Early fusion v2 — unconstrained best AUC, but text 96% of the decision	76.6%	70.9%	68.1%	75.2%	0.877	0.634	92.9%	57.6%
Early fusion v2 — shipped configuration L2, all three modalities count	85.8%	72.3%	76.6%	79.2%	0.846	0.686	85.7%	72.7%

[†] Late-fusion training accuracy was recomputed from the saved models on re-extracted corpus features. The text features match the originals exactly; the audio and video features differ slightly, so its test accuracy reproduces exactly (70.2%) and its test AUC to within

0.003 (0.766 vs 0.768 recorded). The table shows the recorded test values. Early fusion v1's official run used a model search that was not saved, so its training and CV accuracy are not available.

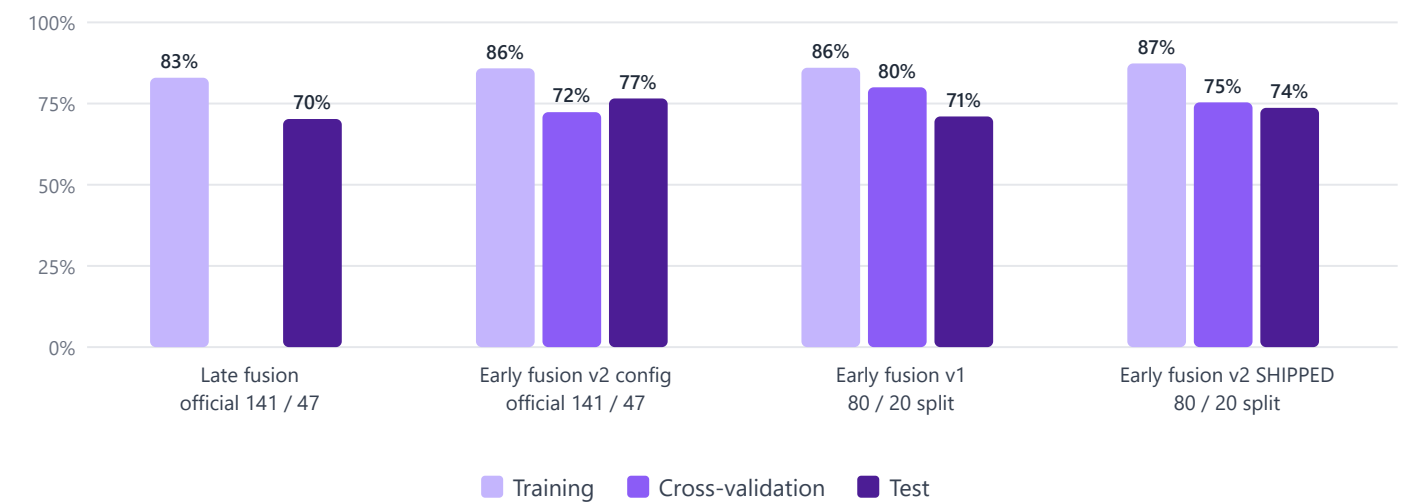
Figure 1 — Held-out test quality on the official split (47 people)



3. The shipped model on its own split (80 / 20)

Model	Training accuracy	5-fold CV accuracy	Test accuracy	Test balanced accuracy	Test ROC-AUC	Test F1	Sensitivity	Specificity
Early fusion v1 packaged model	86.0%	80.0%	71.1%	74.2%	0.859	0.621	81.8%	66.7%
Early fusion v2 — SHIPPED MODEL the model serving on the website	87.3%	75.3%	73.7%	68.0%	0.795	0.545	54.5%	81.5%

Figure 2 — Training, cross-validation and test accuracy



4. Previous late-fusion system, model by model (official split)

Model	Training accuracy	Test accuracy	Test ROC-AUC	Test F1
Text model	81.6% [†]	70.2%	0.753	0.562
Audio model	73.8% [†]	57.4%	0.602	0.444
Video model	68.8% [†]	57.4%	0.584	0.474
Late fusion (all three)	83.0% [†]	70.2%	0.768	0.611

Late fusion trained a separate model per modality and combined their three probabilities. Audio and video on their own were close to chance on the test set; text carried the system.

5. Where the test errors are

Late fusion — official test (47)			Early fusion v2 config — official test (47)			Early fusion v2 shipped — 80/20 test (38)		
	Predicted not depressed	Predicted depressed		Predicted not depressed	Predicted depressed		Predicted not depressed	Predicted depressed
Actually not depressed	22	11	Actually not depressed	24	9	Actually not depressed	22	5
Actually depressed	3	11	Actually depressed	2	12	Actually depressed	5	6
accuracy 70.2% · found 11 of 14 depressed			accuracy 76.6% · found 12 of 14 depressed			accuracy 73.7% · found 6 of 11 depressed		

6. What the results show

- **Early fusion beats the previous late fusion on the same test people.** On the official split the v2 configuration reaches 76.6% test accuracy against 70.2% (+6.4 points), ROC-AUC 0.846 against 0.768 (+0.078), and finds 12 of 14 depressed participants against 11.
- **Training accuracy is higher than test accuracy, as expected.** The shipped model scores 87.3% on its training people and 73.7% on unseen ones (a 13.6-point gap; late fusion: 12.8 points). With 150 people and 3,384 measurements some gap is unavoidable. L2 regularisation and compression to 40 PCA components are what keep it bounded, and the cross-validation accuracy (75.3%) agrees with the test result.
- **All three modalities count in the shipped model.** Share of the decision: text 76.2%, audio 12.6%, video 11.2%. The unconstrained variant scores a higher AUC (0.877) but is 96% text; the shipped configuration was chosen so the decision is genuinely multimodal.
- **The test sets are small.** One person is 2.6 accuracy points on the 38-person test and 2.1 on the 47-person official test, so differences of a few points are within noise. The ranking by ROC-AUC, which does not depend on a cut-off, is the steadier comparison.

7. Verification

Every number was recomputed from the saved models and features on this machine and compared with what the training machine recorded.

Check	Recorded	Recomputed	Result
Early fusion v2 shipped — test AUC (80/20)	0.7946	0.7946	exact match
Early fusion v2 shipped — test accuracy (80/20)	0.7368	0.7368	exact match
Early fusion v2 shipped — cross-validation AUC	0.7056	0.7056	exact match
Early fusion v1 — test AUC (80/20)	0.8586	0.8586	exact match
Early fusion v2 config — official test AUC	0.8463	0.8463	exact match
Early fusion v2 config — official test accuracy	0.7660	0.7660	exact match
Late fusion — text model test AUC	0.7532	0.7532	exact match
Late fusion — test accuracy	0.7021	0.7021	exact match
Late fusion — test AUC	0.7684	0.7662	close (see note †)

Sources: EarlyFusion_Work_V2/results (v2_official_protocol.csv, bundle metrics), EarlyFusion_Work/results (official_protocol.csv), My_Work/results (test_results.csv), and the saved model bundles. Aggregate figures only.